

Tilings

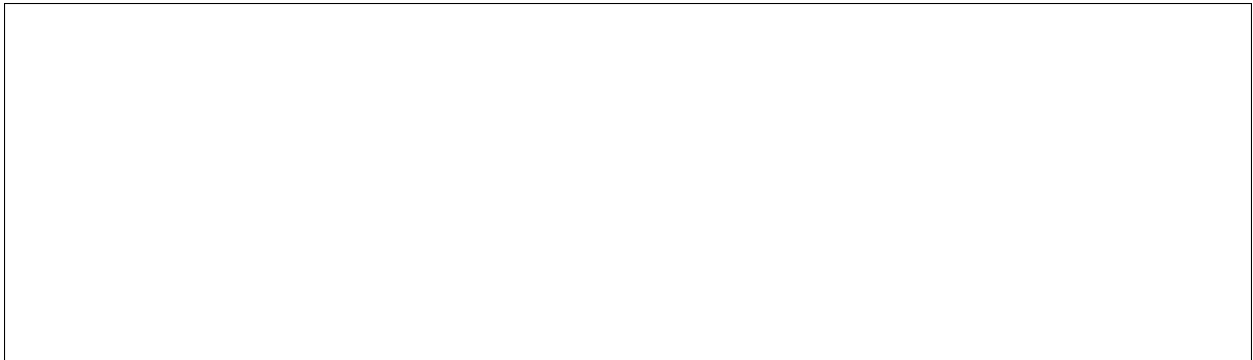
Why only three shapes fill the floor

The question. A tiling covers the floor with no gaps and no overlaps. If every tile is the *same* regular polygon, which shapes can do it? Try it first, then prove which ones must work.

Build: ring a single point

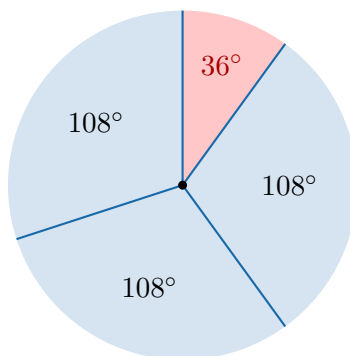
Cut out a stack of identical regular triangles, squares, pentagons, and hexagons (or use pattern blocks). For each shape, fit copies around one point, corner to corner, and see whether they close up with no gap and no overlap.

Which shapes closed up cleanly? Which left a gap or overlapped? Draw your best pentagon attempt and circle the gap:



The key idea

At any meeting-point, the corners crowding around it must sweep the *whole* turn: they add up to exactly 360° . So a regular shape tiles only when copies of its corner angle total 360, which means its corner angle must **divide** 360 **evenly**. The corner angle of a regular n -gon is $\frac{(n-2) \cdot 180}{n}$ (you proved the $(n-2) \cdot 180$ part in Module 02).



Three pentagon corners reach only 324; a 36° gap is left, and a fourth pentagon will not fit.

Fill in the table

Compute each corner angle, then decide. Use $\frac{(n-2) \cdot 180}{n}$.

shape	n	corner = $\frac{(n-2)180}{n}$	divides 360?	tiles?
triangle	3			
square	4			
pentagon	5			
hexagon	6			
heptagon	7			

In one sentence: why can no regular polygon with 7 or more sides ever tile? (Hint: how big is its corner, and how many could meet at a point?)

Push further

1. **Mix the shapes (semiregular tilings).** Allow *different* regular polygons at a vertex, as long as the angles still total 360 and every vertex looks the same. Fill in the missing angle:

- square + two octagons (octagon corner = 135): $90 + 135 + \underline{\hspace{2cm}} = 360$.
- triangle, hexagon, triangle, hexagon: $60 + 120 + 60 + \underline{\hspace{2cm}} = 360$.
- four triangles + one hexagon: $60 \times 4 + \underline{\hspace{2cm}} = 360$.

There are exactly **eight** semiregular tilings. How many can you find?

2. **Competition gem.** Which multisets of regular-polygon corner angles add to exactly 360 at a point? It is a *finite* hunt (a big-angle polygon leaves little room). For example 3.4.4.6 works: $60 + 90 + 90 + 120 = 360$. List a few more, then ask the harder question: which of them actually tile the whole floor? (Not all do.)

3. **True or false?** A big enough regular polygon will eventually tile the plane all by itself. T / F